



Customer Shopping Behavior Analysis

Uncovering insights from **3,900 transactions** across product categories — analyzing spending patterns, customer segments, product preferences, and subscription behavior to guide strategic decisions.

Dataset at a Glance



Key features include demographics, purchase details, and shopping behavior — from discount usage to shipping type.

3,900 Rows

Transactions analyzed

18 Columns

Features per record

37 Missing

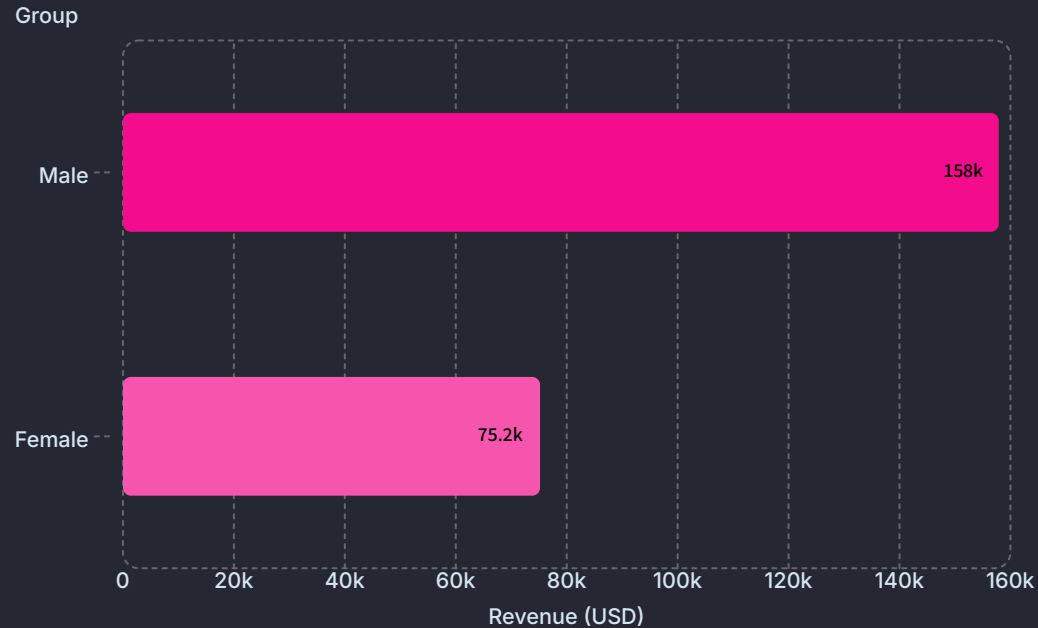
Review Rating values imputed

Data Preparation in Python



Columns were standardized to snake_case, redundant fields dropped (promo_code_used), and the cleaned DataFrame was pushed to PostgreSQL for SQL analysis.

Revenue by Gender & Shipping



Shipping Type vs. Avg. Spend

Male customers generate **2× more revenue** than female.
Express shipping users spend slightly more on average.

Standard Shipping

\$58.46 avg. purchase

Express Shipping

\$60.48 avg. purchase

Top-Rated Products & Discount Leaders

Top 5 Products by Avg. Rating

Product	Avg. Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Most Discount-Dependent Products

Product	Discount Rate
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Customer Segmentation

3,116

Loyal

Customers with strong purchase
history

701

Returning

Moderate repeat buyers

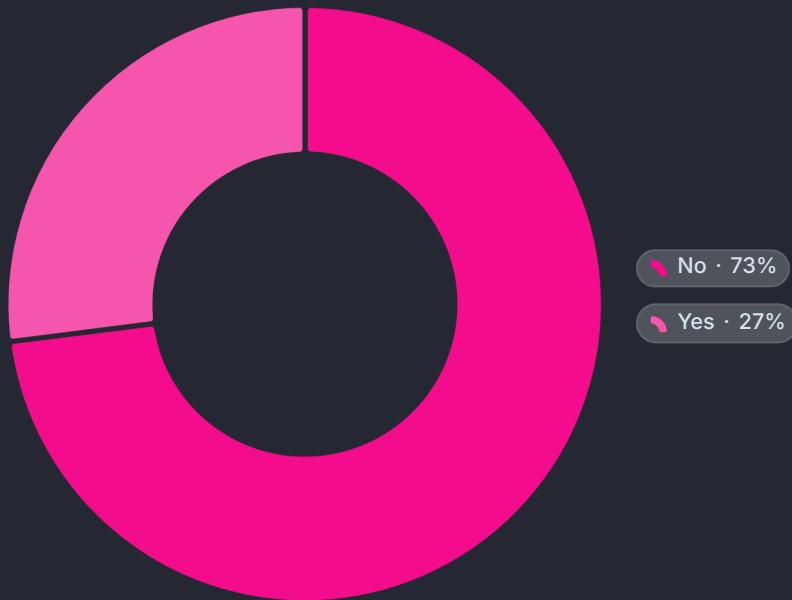
83

New

First-time or recent customers



Subscriptions & Repeat Buyers

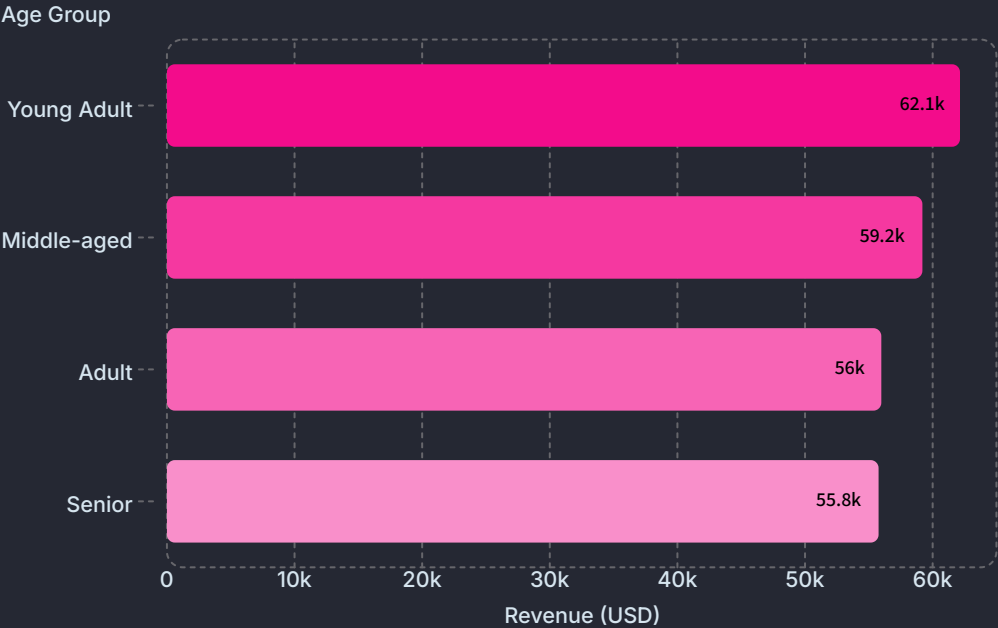


Subscriber vs. Non-Subscriber Spend

Status	Avg. Spend	Total Revenue
Subscribers	\$59.49	\$62,645
Non-Subscribers	\$59.87	\$170,436

Only **27% of customers subscribe**. Of repeat buyers (>5 purchases), 958 subscribe vs. 2,518 who don't — a major growth opportunity.

Revenue by Age Group & Top Products by Category



Top Products per Category

Category	#1 Product	Orders
Accessories	Jewelry	171
Clothing	Blouse	171
Footwear	Sandals	160
Outerwear	Jacket	163



Power BI Dashboard

3.9K

Customers

\$59.76

Avg. Purchase

3.75

Avg. Rating

Interactive filters by subscription status, gender, category, and shipping type. Visualizations include revenue by category, sales by age group, and subscription breakdown.

Business Recommendations



Boost Subscriptions

Promote exclusive benefits to convert the 73% non-subscribers.



Loyalty Programs

Reward repeat buyers to grow the Loyal segment.



Review Discount Policy

Balance sales boosts with margin control on high-discount items.



Targeted Marketing

Focus on Young Adults and express-shipping users — highest revenue contributors.